

THIVYA DHANASEGARAN

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WORK EXPERIENCE

Lead Teaching Assistant – Northeastern University, Boston, MA Jan 2025 – Present

- Assisted students in implementing efficient ADTs and data structures like Stacks, Queues, Lists, and Trees
- Addressed student queries, designed course materials, and collaborated on assignments and exams

Teaching Assistant – Northeastern University, Boston, MA Sep 2024 – Dec 2024

- Assisted students with HTML, CSS, JavaScript, Node.js, Express, React, TypeScript, MongoDB, and RESTful APIs
- Graded 100+ assignments, provided feedback, conducted office hours, and developed course materials

Senior Systems Engineer – Infosys Limited, India – 2.8 years

Project: Employee Resource Management System

- Developed a **Java Spring MVC** based Employee Resource Management System with **Hibernate** and **MySQL**
- Boosted system performance by 40% and reduced database load by 30% by integrating **Redis caching**
- Strengthened security with role-based access control via **Spring Security**, ensuring proper user access and data protection
- Improved issue resolution time by 30% through effective teamwork, communication, and problem-solving skills

Systems Engineer – Infosys Limited, India – 2.5 years

Project: MNCR Prospective Conversion, Client: LexisNexis

- Automated conversion of unstructured legal documents into MNCR XML format using **Informatica DT/DX**, decreasing manual processing time by 50%, and received positive feedback from the client for enhancing operational efficiency

EDUCATION

Northeastern University, Boston, MA – Master of Science in Information Systems CGPA: 3.77 | Expected April 2025

ACADEMIC PROJECTS

Network Structures and Cloud Computing | [GitHub](#) Jan – Apr 2024

Tech Stack: GCP, Terraform, GitHub Actions, Packer, Node.js, Express, Sequelize ORM, MySQL, Jest

- Developed and deployed cloud-native backend applications on GCP using Node.js, Express, and MySQL
- Provisioned custom VPCs, Cloud SQL, VMs with Terraform as IaC and automated machine image creation with Packer
- Optimized reliability with load balancers, autoscaling, Cloud Functions, and real-time email notifications through Pub/Sub
- Streamlined deployments with CI/CD pipelines using GitHub Actions and managed version control with GitHub

NEU Student Healthcare Management System | [GitHub](#) July – Aug 2024

Tech Stack: Java, Spring Boot, Hibernate, MySQL, Redis, JUnit, Mockito, Spring Security, SLF4J, OAuth, AJAX

- Built a full-stack healthcare management system with modules for health records, appointment scheduling, and prescription handling using Spring Boot, Hibernate, MySQL, and REST APIs
- Enhanced database performance by 40% with Redis caching, automated email notifications, and unit testing using JUnit
- Secured the application with Google OAuth, role-based access control, and deployed using Docker for seamless scalability

Campsite Booking Web Application | [GitHub](#) Nov – Dec 2023

Tech Stack: TypeScript, React.js, Node.js, Express.js, MongoDB, Redux, Rest API, Swagger, i18n

- Designed a responsive campsite booking web application using the MERN stack and REST API
- Enabled users to search, book campsites, manage their own campsites, and add reviews
- Used Node.js/Express.js for backend services, MongoDB for data storage, and i18n for multi-language support

Language Learning Platform | [GitHub](#) Jan – Apr 2024

Tech Stack: SQL, Data Modeling, Normalization, Triggers, Procedures, SQL Server 2022, Power BI, Azure Data Studio

- Created Language Learning System database by designing ER diagrams and applying normalization principles
- Fine-tuned database performance using stored procedures, triggers, views, and indexes
- Secured sensitive user data with AES-256 encryption and generated insightful reports using Power BI

Prompt Engineering for Generative AI Jan 2025 – Present

Tech Stack: Prompt Design & Optimization, GPT models, Hugging Face Transformers, LLMs Fine-Tuning

- Learned prompt creation and fine-tuning AI models for diverse applications such as content creation and decision-making
- Gained experience in training and optimizing AI models with reinforcement learning for enhanced adaptability

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, TypeScript, SQL, Algorithm & Data Structure, Object-Oriented Programming

Web Technologies: Spring Boot, React, Node.js, Express.js, Servlets, HTML, CSS, REST APIs, Spring MVC, Redux, AJAX

Database: MySQL, MongoDB, Redis, Postgres, Azure Data Studio, ORM (Hibernate, Mongoose, Sequelize)

Cloud: GCP, Terraform, Jenkins, GitHub Actions (CI/CD), Docker, Packer, Microservices

Others: GitHub, Linux, Agile SDLC, jQuery, Postman, Maven, JSON, RabbitMQ, Swagger, GraphQL, Elasticsearch